

# Who Pays the Bill?

Substrate Skepticism and the Memory–Innovation–Counterfactual Frontier

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## Abstract

This note asks a deliberately modest question. Suppose the observable world is described only through controlled stochastic input–output laws. Without claiming to identify the underlying ontology, what information resources must *some* exact realization possess? We call the resulting stance *substrate skepticism*: the substrate is left unspecified, while the admissible realization contract is made explicit and its resource cost is studied.

The note is intentionally written under a strong priority convention: *we assume a priori that some or all of the mathematics below has already been developed under other terminology, unless and until a careful literature search establishes otherwise*. Our motive is amateur curiosity, not a claim of priority. In fact, much of the surrounding territory is already classical or recently active: minimum-entropy coupling, random-number generation, randomness recycling, random-map realizations, mixed versus behavioral strategies, catalytic entropy, counterfactual MDP couplings, and causally stable common random numbers.

The main technical attack isolates a very small test problem. A memoryless controlled channel has finitely many actions  $a \in \mathcal{A}$  and exact output laws  $P_a$ . An on-path sampler, which is allowed to generate randomness only after the chosen action is known, is already covered very strongly by modern online randomness-recycling results and can asymptotically approach the Shannon lower bound. A stronger *counterfactual* contract asks for one latent state that simultaneously defines the responses to all actions and can be reused over time. For a particularly strict “reset” version of this contract we prove an elementary cancellation result: if two non-permutation output distributions have the same Shannon entropy, then no finite discrete reservoir can operate exactly at the Shannon entropy floor while resetting independently after every step. The obstruction is not Shannon entropy itself but the full probability spectrum; equality at the Shannon floor forces an entropy-preserving, hence essentially invertible, transformation, and tensor-product cancellation then preserves all power sums.

This does *not* establish a positive asymptotic counterfactual entropy tax. Rather, it reveals the next genuinely interesting questions: whether the Shannon floor is an unattained infimum, how the excess rate decays with available memory, whether hidden cross-counterfactual correlations can recycle the spectral mismatch, and which notion of memory remains meaningful once infinite precision is allowed. The resulting object is not a single “bill”, but a memory–innovation–precision frontier conditioned on an explicit accounting standard.

*“We found the bill. We did not find the accountant.”*

## 1 Priority convention and philosophical origin

### 1.1 The priority convention

The purpose of this note is to sharpen questions, not to manufacture novelty. We therefore adopt the following rule throughout.

**Priority convention.** Assume that any clean statement discovered here has probably been stated, proved, or embedded in a more general theory somewhere else. A

statement is called *derived here* only to describe the internal logic of this note; the phrase does not mean historically new. When a nearby or stronger precedent is known, it is cited explicitly. When no direct precedent has been located, we say only that we have not located one.

This convention is not rhetorical modesty. It is technically necessary. The problem touches several mature literatures whose terminology differs sharply despite substantial mathematical overlap. Minimum-entropy coupling is established information theory [2]. Random-map realizations of stochastic dynamics are classical in probability and simulation; entropy-efficient random-number generation has a long history [3, 4]; recent work gives near-Shannon online sampling by explicitly recycling unused randomness [5, 6]. Extensive-form game theory has long distinguished ex-ante randomization over complete contingent plans from local behavioral randomization [8, 9], with modern finite-memory refinements [10]. Catalytic information theory and resource theories show that correlations and auxiliary systems can radically change which exact or approximate state transformations are possible [13, 14]. Recent reinforcement-learning work explicitly separates per-action marginals from joint counterfactual couplings [11], and recent simulation methodology has emphasized that merely reusing a PRNG seed need not provide causally coherent cross-intervention noise [12].

Accordingly, the most important deliverable here is not a new name. It is a clean map of which question is being asked under which realization contract.

## 1.2 From Pavel Arazim’s answer to substrate skepticism

A philosophical trigger for the present line of thought was Pavel Arazim’s answer for the Czech project *Zeptej se vědce* to the question “*Co když je život sen a když člověk umře, tak se probudí?*” [1]. The answer distinguishes existential from theoretical skepticism and, in its theoretical part, warns against expecting a final argument that identifies an ultimate reality behind ordinary experience.

The present note does *not* claim that Dr. Arazim endorses any of the mathematical framework below. The connection is only motivational. Instead of asking

“What is the substrate?”

we ask a weaker question:

If an exact substrate exists under a stated realization contract, what resources must it carry?

This shift is what we call *substrate skepticism*. It refuses to infer ontology from an implementation bound, but it also refuses the opposite mistake of treating exact realization as resource-free.

The original public answer is available at

[https://zeptejsevedce.cz/dotazy-a-odpovedi/  
co-kdyz-je-zivot-sen-a-kdyz-ctovek-umre-tak-se-probudi/](https://zeptejsevedce.cz/dotazy-a-odpovedi/co-kdyz-je-zivot-sen-a-kdyz-ctovek-umre-tak-se-probudi/).

## 2 The accounting-standard principle

A recurring failure mode in discussions of simulation, randomness, or “fundamental reality” is to move between different implementation contracts without noticing. The same observable law can have very different resource costs depending on what the simulator is required to make jointly definite.

We will distinguish three levels.

**C1. Chosen-path sampling.** At time  $t$ , the action  $A_t$  is already known. The simulator need only produce the actually requested output  $Y_t \sim P_{A_t}$  with the correct conditional law.

**C2. One-step counterfactual table.** Before an action is selected, a single latent variable determines a vector

$$F_t = (F_t(a))_{a \in \mathcal{A}}$$

whose coordinate marginals are  $F_t(a) \sim P_a$ . Only one coordinate is observed, but all coordinates are jointly defined.

**C3. Reusable counterfactual substrate.** The latent information that generated the current counterfactual table is not simply discarded. Some reservoir survives to the next step and must remain valid for all future adaptive action choices. This is the contract most closely related to a single coherent response tree.

The distinction between C1 and C2 is already explicit in modern joint-MDP language: an ordinary MDP fixes action-conditioned marginals but not a joint coupling of the potential outcomes under several actions [11]. C3 is stronger still because it constrains how the joint counterfactual structure persists through time.

This yields our first principle:

**The bill is meaningful only after the accounting standard is fixed.**

### 3 A minimal controlled channel

Let  $\mathcal{A}$  and  $\mathcal{Y}$  be finite. A memoryless controlled channel is a family

$$W = \{P_a : a \in \mathcal{A}\}, \quad P_a \in \Delta(\mathcal{Y}).$$

For every action sequence, including an adaptive one,

$$\Pr(Y_t = y \mid A_{1:t}, Y_{1:t-1}) = P_{A_t}(y). \quad (3.1)$$

An admissible policy may choose

$$A_t = \pi_t(A_{1:t-1}, Y_{1:t-1}).$$

Thus the policy may probe any correlation that becomes visible through the realized history, but it cannot access private simulator state directly.

Define the largest one-step observable entropy

$$h_{\max} := \max_{a \in \mathcal{A}} H(P_a).$$

Any exact simulator must supply at least  $h_{\max}$  bits per step along the policy that always selects an entropy-maximizing action, in the appropriate asymptotic sense. This lower bound is elementary and says nothing about counterfactual coherence.

### 4 One-shot counterfactual cost: minimum-entropy coupling

Under contract C2, one latent draw  $F = (F(a))_{a \in \mathcal{A}}$  must have the prescribed coordinate marginals. Therefore its minimum possible entropy is exactly the multi-marginal minimum-entropy coupling:

$$C_1(W) := \min \left\{ H(F) : \text{Law}(F(a)) = P_a \ \forall a \in \mathcal{A} \right\} = \text{MEC}(P_a : a \in \mathcal{A}). \quad (4.1)$$

This is not a new construction. Minimum-entropy coupling is an established object [2]; related common-randomness and functional-representation questions have been extensively studied.

The one-shot counterfactual surcharge is

$$r_1(W) := C_1(W) - h_{\max} \geq 0. \quad (4.2)$$

It is tempting to interpret  $r_1$  as a per-step tax. That temptation is exactly what we will attack.

## 5 The chosen-path attack: most of the rate disappears

Suppose the simulator is required only to satisfy C1. Then modern exact online sampling already comes remarkably close to a complete answer.

Draper and Saad give randomness-recycling algorithms that generate arbitrary stochastic sequences of discrete random variables with amortized entropy consumption arbitrarily close to the Shannon lower bound while maintaining only logarithmic persistent state in the accuracy parameter [5]. Draper, Harris, and Saad subsequently treat real-valued probabilities and obtain, for a sequence of target distributions  $F_1, F_2, \dots$ , an exact sampler with expected coin consumption

$$\mathbb{E}[T_n] \leq \mathbb{E} \left[ \sum_{t=1}^n H(F_t) \right] + O(\log n), \quad (5.1)$$

using  $O(\log n)$  persistent bits in their model [6]. Their formulation explicitly allows each target distribution to depend on previous outputs, provided it does not depend on private sampler state once the public history is fixed. That is exactly the information pattern of an adaptive controller choosing  $P_{A_t}$  from the observed history.

Thus, under C1, the broad claim

“adaptive choice itself forces a positive entropy-rate surcharge”

is untenable. The actually requested stochastic output can be generated at asymptotically Shannon-optimal entropy cost.

Recent lower bounds make the complementary point: getting arbitrarily close to the Shannon rate is not free in memory. Draper and Saad show that exact samplers that achieve rate  $H + \varepsilon$  uniformly over broad classes require persistent space growing at least on the order of  $\log(1/\varepsilon)$  [7].

This is already a nontrivial memory-innovation tradeoff, but it is a *chosen-path* tradeoff. It does not by itself price a simultaneously defined counterfactual table.

## 6 A strict reusable-reservoir contract

To isolate the genuinely counterfactual part, consider the following deliberately strong model.

**Definition 6.1** (Strong reset recycler). *Fix a discrete reservoir law  $\lambda$  on a finite state space  $\mathcal{S}$  and a discrete innovation law  $\nu$  on a finite alphabet  $\mathcal{U}$ . A strong reset recycler for  $W$  consists of deterministic maps*

$$\Phi_a : \mathcal{S} \times \mathcal{U} \rightarrow \mathcal{Y} \times \mathcal{S}, \quad a \in \mathcal{A},$$

*such that, for independent*

$$S \sim \lambda, \quad U \sim \nu,$$

*we have*

$$\Phi_a(S, U) = (Y, S') \sim P_a \otimes \lambda \quad \forall a \in \mathcal{A}. \quad (6.1)$$

*Thus the chosen output has exactly the desired law and the reservoir is returned with exactly the same distribution, independently of that output.*

The independence in (6.1) is intentionally stronger than mere observable correctness. It says that after every use the reservoir is a fresh-looking copy of itself, not merely a hidden state whose correlations happen to remain inaccessible to the realized policy. We impose it because it produces a sharp crash-test. Later we will remove it again.

**Proposition 6.2** (Entropy balance). *Every strong reset recycler satisfies*

$$H(\nu) \geq \max_{a \in \mathcal{A}} H(P_a). \quad (6.2)$$

*Proof.* For any action  $a$ , the output pair is a deterministic function of  $(S, U)$ , hence

$$H(Y, S') \leq H(S, U).$$

Using the independence assumptions on both sides,

$$H(P_a) + H(\lambda) \leq H(\lambda) + H(\nu),$$

which gives (6.2).  $\square$

The lower bound is exactly the observable Shannon floor. The interesting question is whether it can be attained.

## 7 The spectrum-cancellation obstruction

Here the small problem becomes unexpectedly rigid.

**Theorem 7.1** (Exact floor forces spectral equivalence). *Let  $P$  and  $Q$  be finite probability vectors satisfying*

$$H(P) = H(Q) = h.$$

*Suppose there exists a finite strong reset recycler with reservoir law  $\lambda$  and innovation law  $\nu$  satisfying*

$$H(\nu) = h$$

*for both actions  $P$  and  $Q$ . Then  $P$  and  $Q$  have the same multiset of nonzero atom probabilities; equivalently, they differ only by a permutation and possible zero padding.*

*Proof.* For the action with target law  $P$ , entropy balance is tight:

$$H(P) + H(\lambda) = H(\nu) + H(\lambda).$$

Since  $(Y, S') = \Phi_P(S, U)$  is a deterministic function of  $(S, U)$ , equality of the input and output entropies gives

$$H(S, U \mid Y, S') = 0.$$

Hence the deterministic map is invertible almost surely on the support. Therefore the atom multiset of

$$\lambda \otimes \nu$$

coincides with the atom multiset of

$$P \otimes \lambda.$$

The same argument for  $Q$  gives

$$P \otimes \lambda \equiv_{\text{perm}} Q \otimes \lambda. \quad (7.1)$$

For every integer  $r \geq 2$ , equality of the product atom multisets implies equality of their  $r$ th power sums:

$$\left( \sum_i p_i^r \right) \left( \sum_j \lambda_j^r \right) = \left( \sum_i q_i^r \right) \left( \sum_j \lambda_j^r \right).$$

The second factor is strictly positive, hence

$$\sum_i p_i^r = \sum_i q_i^r \quad \forall r \geq 2. \quad (7.2)$$

After padding the shorter vector by zeros, finitely many power sums determine the elementary symmetric polynomials through Newton's identities, and therefore determine the multiset of atoms. Thus  $P$  and  $Q$  are permutations of one another.  $\square$

*Remark 7.2* (What is and is not being claimed). The theorem is an elementary cancellation argument, not a historical novelty claim. It is closely aligned with known catalytic-entropy reasoning. In particular, reversible catalytic transformations preserve total entropy, while correlations can account for strict entropy increase; at equality there is no mutual-information budget available to hide a spectral mismatch [13, 14]. We have not located this exact recycler formulation in the literature, but the mechanism should be regarded as belonging to that established neighborhood.

The result is conceptually useful because it separates two resources:

$$\text{Shannon amount} \quad \text{versus} \quad \text{probability spectrum.}$$

Equal Shannon entropy does not mean one-step exact recyclability at the entropy floor.

## 8 The smallest numerical crash-test

Consider

$$P = \left( \frac{1}{2}, \frac{1}{4}, \frac{1}{4} \right)$$

and

$$Q = (q, q, 1 - 2q),$$

where  $q \in (1/3, 1/2)$  is the unique solution of

$$-2q \log_2 q - (1 - 2q) \log_2 (1 - 2q) = \frac{3}{2}. \quad (8.1)$$

Numerically,

$$q \approx 0.4100324281819856,$$

so

$$H(P) = H(Q) = 1.5 \text{ bits.}$$

The two probability spectra are manifestly not permutations.

A direct enumeration of the vertices of the  $3 \times 3$  transportation polytope gives a minimum-entropy coupling with matrix

$$\begin{pmatrix} \frac{1}{2} - q & q & 0 \\ 2q - \frac{3}{4} & 0 & 1 - 2q \\ \frac{1}{4} & 0 & 0 \end{pmatrix}, \quad (8.2)$$

whose entropy is

$$\text{MEC}(P, Q) \approx 2.05391858560726 \text{ bits.} \quad (8.3)$$

Hence the one-shot counterfactual surcharge is

$$r_1 = \text{MEC}(P, Q) - 1.5 \approx 0.55391858560726 \text{ bits.} \quad (8.4)$$

By the spectrum-cancellation theorem, the strong reset model cannot operate with exactly 1.5 fresh bits of Shannon entropy per step using any finite discrete reservoir and finite discrete innovation law. Yet the one-shot stateless upper bound is only about 2.0539 bits. Thus the tiny example creates a genuine interval:

$$1.5 \quad (\text{Shannon floor, unattainable under exact finite reset}) \quad < \quad R \quad \leq \quad 2.0539185856. \quad (8.5)$$

The unresolved issue is whether the left endpoint is merely unattained or is separated from the feasible region by a positive asymptotic gap once a concrete finite-memory model is fixed.

## 9 Near equality implies near reversibility

The previous proof has a useful quantitative reformulation.

Suppose a strong reset recycler uses innovation entropy

$$H(\nu) = h + \delta$$

for a target  $P$  with  $H(P) = h$ . Let

$$Z = (S, U), \quad W = (Y, S').$$

Because  $W$  is a deterministic function of  $Z$ ,

$$H(Z | W) = H(Z) - H(W) = \delta. \quad (9.1)$$

Thus every near-Shannon strong recycler must be *nearly reversible in conditional entropy*. The excess fresh entropy is exactly the average information about the input reservoir–innovation pair that is irreversibly merged by the update.

For two actions  $P, Q$  of equal entropy, the same source law  $\lambda \otimes \nu$  admits deterministic coarse-grainings to both  $P \otimes \lambda$  and  $Q \otimes \lambda$ , each losing only  $\delta$  bits. In the coupling-entropy pseudometric of [2], this implies the schematic bound

$$d_{\text{CE}}(P \otimes \lambda, Q \otimes \lambda) \leq 2\delta, \quad (9.2)$$

using the common source as the intermediate coupling. Therefore the small- $\delta$  problem can be reframed as a catalytic approximation question:

$$\boxed{\inf_{\lambda} d_{\text{CE}}(P \otimes \lambda, Q \otimes \lambda) \stackrel{?}{=} 0.} \quad (9.3)$$

We do not know the answer in the exact operational model relevant here. Equation (9.3) should be read as a guide for further attack, not as a claim that a new metric has been discovered. Catalytic majorization and correlated catalysis already show how auxiliary distributions can erase obstructions that are visible in one-shot spectra [13, 14].

## 10 Why the reset condition may be too strong

The strong reset condition

$$(Y_t, S_{t+1}) \sim P_{A_t} \otimes \lambda$$

forces the surviving reservoir to be independent of the public output after each use. A coherent response tree does not obviously require this.

The reservoir might remain correlated with the past in a way that no single realized policy can observe. Such information can live entirely in relations among unchosen counterfactual branches. It is neither ordinary predictive memory nor ordinary public randomness. It is closer to a *counterfactual reserve* or *gauge memory*: information that matters only for how mutually exclusive interventions are embedded into one common latent world.

This possibility matters because correlated catalysis is known to be much more powerful than uncorrelated catalysis. In classical and quantum resource theories, allowing the catalyst to retain its marginal state while building correlations with the transformed system can collapse large families of constraints down to ordinary entropy inequalities in appropriate settings [13, 14]. Related work explicitly treats uncorrelatedness itself as a consumable catalytic resource [15].

Therefore the spectrum-cancellation theorem must not be overinterpreted. It proves a sharp obstruction for one accounting standard. It does not prove a universal entropy-rate tax for all coherent substrates.

## 11 The infinite-precision loophole

Any claim about “fresh randomness” is meaningless unless precision is priced.

Let

$$U \sim \text{Unif}[0, 1].$$

Up to the null set of dyadic ambiguities, the binary expansion

$$U = 0.B_1B_2B_3 \dots$$

contains an infinite i.i.d. fair-bit tape. One real-valued initial condition can therefore encode arbitrarily many future random bits. Equivalently, interval algorithms and arithmetic-decoding viewpoints treat unused location within a probability interval as reusable randomness [3, 6].

If a continuous state is charged only by differential entropy, this creates nonsense accounting: the uniform variable has differential entropy zero in bits relative to Lebesgue measure, yet it carries unbounded discrete precision. Hence a substrate resource theory that permits exact real numbers must include a precision or description-complexity coordinate.

This gives another hard rule:

**A finite Shannon-entropy bound is not a precision bound.**

The distinction is not philosophical decoration. It determines whether “the simulator preloaded all future randomness in one real number” is a free trick or an infinitely expensive one.

## 12 A memory–innovation–precision frontier

The previous sections suggest that a single number is the wrong object. We propose a family of resource regions conditioned on an explicit realization contract.

For a contract  $C$ , let

$$\mathcal{F}_C(W) \subseteq [0, \infty]^3$$

denote the set of achievable triples

$$(M, R, \Pi),$$

where, schematically,

- $M$  measures persistent latent memory,
- $R$  measures fresh innovation consumption per step,
- $\Pi$  measures precision or representational complexity.

The exact definitions must depend on the computational model. For a finite discrete machine one may use  $\log_2 |S|$  or  $H(S)$  for memory and expected consumed fair bits for innovation. For exact real computation, a bit-complexity or oracle-precision model is unavoidable.

A one-dimensional projection already exists in the online-sampling literature: near-optimal entropy consumption requires increasing persistent memory, with upper and lower bounds of logarithmic type in the entropy slack [5, 6, 7]. What remains unclear is the additional deformation of this frontier when the simulator must maintain simultaneous counterfactual structure rather than only the chosen path.

This is the central technical question of the note:

How much larger is  $\mathcal{F}_{\text{counterfactual}}(W)$  than  $\mathcal{F}_{\text{on-path}}(W)$ ?

(12.1)

## 13 Counterfactual breadth

There is another parameter that a one-path sampler does not see: the width of the intervention family that must remain jointly meaningful.

For a finite family of policies  $G$ , earlier finite-horizon response-tree calculations compare an unrestricted multi-policy coupling cost  $\kappa_n^G$  with a tree-compatible cost  $\chi_n^G$ . For fixed finite  $G$ , full-history no-remerging means that a particular pair of policies stops sharing tree nodes after its first action disagreement. This tends to make long-run coherence penalties transient in rapidly mixing systems.

A physical or metaphysical “world valid under any intervention”, however, corresponds more naturally to the entire class

$$\mathcal{G}_n = \{\text{all deterministic nonanticipative policies up to time } n\}.$$

Now new policy pairs continue to share longer and longer prefixes. Coherence constraints are regenerated at every depth of the tree.

This motivates a second resource axis:

$$B_n := \log |\mathcal{G}_n|,$$

or, more structurally, the number of independently branching intervention sites that must be supported. A possible phase-transition question is whether an extensive counterfactual cost can appear only when the policy family itself grows exponentially with horizon.

Recent joint-MDP work deliberately restricts dependence to one-step counterfactual outcomes and then refreshes future exogenous noise independently across branches, in part to avoid the exponential full-tree object [11]. This makes it an especially useful neighboring model: it shows that “full counterfactual tree” is an additional modeling commitment, not something already encoded by an ordinary MDP.

## 14 Causal event identity and the meaning of “the same randomness”

Even if one grants a common exogenous tape, another ambiguity remains: which draw in one intervention world corresponds to which draw in another?

Buffalo, Pearson, and Klein point out that simply reusing the same PRNG seed across intervention scenarios can produce causally incoherent comparisons when interventions change execution order. Their proposed remedy is to key randomness to stable event identities rather than to draw indices [12]. This is directly relevant to substrate accounting. A source can be cheap only after we say what counts as the *same* exogenous event across worlds.

Thus counterfactual cost has at least three layers:

$$\text{marginal law} < \text{joint response table} < \text{stable event-indexed response tree}.$$

Different choices need not have the same minimum entropy.

## 15 Attack log: what did not survive

The following candidate conclusions were deliberately attacked and should not be stated as results.

- A1. “Every adaptive exact simulator must pay the one-shot MEC surcharge at every step.”**

No. Under the chosen-path contract, online randomness recycling approaches the Shannon lower bound [5, 6]. The one-shot counterfactual table cost cannot simply be multiplied by time.

**A2. “Equal Shannon entropy is enough for exact finite-state recycling at the entropy floor.”**

No under the strong reset contract. The spectrum-cancellation theorem rules this out unless the target atom spectra agree.

**A3. “Therefore there is a positive asymptotic counterfactual rate tax.”**

Not established. Failure of exact attainment at finite complexity does not imply a positive infimum gap. The gap may tend to zero as reservoir size, precision, or allowed correlation grows.

**A4. “A finite amount of initial randomness cannot generate an arbitrarily long random world.”**

True for a finite-entropy discrete seed, false if infinite real precision is treated as free. A single exact real variable can encode an infinite random tape.

**A5. “The hidden reservoir may be assumed independent of the past without loss of generality.”**

Not proved and probably too strong. Correlated catalytic resources show that hidden correlations can be operationally powerful [13, 14, 15].

The surviving research problem is consequently narrower and, in our view, more interesting.

## 16 Open problems

**Open Problem 16.1** (Attainment at the Shannon floor). *For the concrete pair*

$$P = (1/2, 1/4, 1/4), \quad Q = (q, q, 1 - 2q), \quad H(P) = H(Q) = 1.5,$$

*determine the infimum fresh-entropy rate under an exact reusable-reservoir model that is strong enough to define both counterfactual actions at every step but weak enough to permit hidden correlations across time.*

*Does the infimum equal 1.5? If yes, is it unattainable at every finite memory/precision budget?*

**Open Problem 16.2** (Finite-complexity gap). *Fix finite reservoir and innovation alphabet sizes  $N, K$ . Let*

$$R_{N,K}^*(P, Q)$$

*be the minimum innovation entropy among exact strong reset recyclers with  $|\mathcal{S}| \leq N$  and  $|\mathcal{U}| \leq K$ . Compactness gives attainment. The spectrum-cancellation theorem implies*

$$R_{N,K}^*(P, Q) > 1.5$$

*for the non-permutation equal-entropy example whenever the feasible set is nonempty.*

*Determine the asymptotic behavior of*

$$\Delta_{N,K} := R_{N,K}^*(P, Q) - 1.5$$

*as  $N, K \rightarrow \infty$ .*

**Open Problem 16.3** (Catalytic coupling distance). *Study*

$$d_{\text{cat}}(P, Q) := \inf_{\lambda} d_{\text{CE}}(P \otimes \lambda, Q \otimes \lambda),$$

with the admissible class of catalysts specified explicitly. When  $H(P) = H(Q)$ , must  $d_{\text{cat}}(P, Q) = 0$ ? How does the answer change if the catalyst must be finite, computable, independent at each cycle, or reusable with correlations?

We do not claim that this formulation is new; its relationship to catalytic majorization, correlated catalysis, and entropy-coupling geometry must be established first.

**Open Problem 16.4** (Hidden-synergy recycler). *Can a finite hidden state remain correlated with the observed past while being invisible to every future adaptive policy, yet still carry counterfactual information that lowers future innovation cost?*

If so, characterize the maximal recyclable entropy contained in such observationally inaccessible but counterfactually relevant correlations.

**Open Problem 16.5** (Full-tree versus chosen-path rate). *For a memoryless controlled channel, compare:*

$R_{\text{path}}(W) := \text{minimum asymptotic entropy consumption of an exact sampler valid for every adaptive chosen path}$

$R_{\text{tree}}(W) := \text{minimum entropy rate of one source that fixes a causally coherent response tree for all adaptive paths}$

Known online sampling results strongly constrain  $R_{\text{path}}$  [5, 6]. Determine whether

$$R_{\text{tree}}(W) > R_{\text{path}}(W)$$

can occur for a finite memoryless channel under a natural finite-precision source model.

**Open Problem 16.6** (Breadth transition). *Let  $G_n$  be a policy family growing with horizon. Determine how the asymptotic coherence premium depends on the growth and branching structure of  $G_n$ . Is there a threshold between fixed/subexponential counterfactual families and exponentially branching families at which an extensive premium becomes possible?*

**Open Problem 16.7** (Precision accounting). *Find a representation-invariant resource measure that prevents an exact real number from serving as a free infinite random tape while remaining compatible with ordinary finite-state random sampling. Candidate ingredients include expected bit precision, oracle-query complexity, computable-information constraints, or a rate-distortion relaxation.*

**Open Problem 16.8** (Stable event identity). *Quantify the additional entropy or memory cost of requiring a source to be keyed to stable modeled-event identities across interventions, rather than merely reproducing each intervention law separately. This is the resource version of the causal-coherence issue identified for stateful PRNGs in [12].*

## 17 Substrate skepticism: what survives the mathematics

The mathematics supports a narrow but useful philosophical position.

### 17.1 What it does not tell us

No entropy inequality identifies the substrate. A stochastic law realized by fundamental physics, a simulator, a deeper deterministic dynamics with a huge seed, or a mathematically equivalent representation may agree on all observations. Resource accounting does not prove

fundamentality,

does not disprove

simulation,

and does not by itself decide whether randomness is ontic or epistemic.

## 17.2 What it can tell us

Once an implementation contract is fixed, exactness imposes constraints. A finite discrete seed cannot deterministically generate outputs whose Shannon entropy grows without bound. A strong reusable reservoir operating at the Shannon floor cannot erase probability-spectrum mismatch by magic. A chosen-path simulator can recycle unused randomness, but doing so near optimally requires memory. An infinite-precision initial condition can prepay arbitrarily much future randomness, but only because infinite precision has been admitted as an unpriced resource.

Thus the substrate-skeptical statement is not

“we know what lies underneath.”

It is

“We do not know the payer, but we can constrain the admissible forms of payment.”

## 17.3 From one bill to a resource geometry

The original metaphor of a single bill is too simple. There are exchange rates:

memory  $\longleftrightarrow$  fresh randomness  $\longleftrightarrow$  precision  $\longleftrightarrow$  counterfactual breadth.

A substrate may pay less innovation by storing reusable randomness. It may pay less explicit storage by increasing precision. It may avoid an enormous full response tree by refusing counterfactual definiteness beyond realized interventions. Each move changes the accounting standard.

The mathematically interesting invariant, if one exists, should therefore be something stable under benign recodings but sensitive to these genuine resource substitutions.

## 18 A sharper answer to “Who pays?”

The question

Who pays?

is ontological. This note does not answer it.

But it can refine the question into four operational ones:

- (i) How much entropy must enter along the realized path?
- (ii) How much persistent memory is required to recycle unused randomness?
- (iii) How much latent structure is required to keep mutually exclusive interventions in one coherent response tree?
- (iv) How much precision has been silently treated as free?

The first two already connect directly to an active and rapidly developing random-sampling literature [5, 6, 7]. The third is the least settled component in the formulation used here. The fourth is necessary to prevent continuous-state idealizations from making the first three vacuous.

This is, in our view, the strongest form of substrate skepticism available from the present mathematics:

We do not infer what reality is made of. We specify what an exact realization is required to preserve, then ask for the cheapest admissible implementation. The answer is a constraint on possible substrates, not an identification of the actual one.

## 19 Conclusion

The attack changed the target several times.

The one-shot quantity  $\text{MEC}(P_a : a \in \mathcal{A})$  is real but is not automatically a per-step tax. Chosen-path randomness can be recycled nearly down to the Shannon rate, and modern work already gives sharp space–entropy tradeoffs. A stronger reusable counterfactual contract introduces a different obstruction: at exact Shannon equality, a finite independently resetting reservoir cannot generally absorb a mismatch in probability spectra. The smallest three-outcome example already exhibits this obstruction.

But that is not the end of the accounting. Correlated hidden reservoirs may be stronger. Counterfactual breadth may matter. Infinite precision can masquerade as infinite prepaid randomness. Consequently, the most defensible object is not a scalar “cost of reality” but a contract-dependent frontier of memory, innovation, precision, and counterfactual breadth.

The philosophical lesson is correspondingly restrained:

**The bill is real only after the accounting standard is fixed.**

And the deliberately unanswered final question remains:

**Who pays?**

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The author is not writing from an academic position in information theory, probability, game theory, causal inference, or mathematical physics. The investigation is an amateur one in the literal sense: it is being pursued because the questions are interesting. The standing assumption remains that any elegant theorem encountered along the way probably has a prior owner whom we have not yet found.

## A Reproducibility note for the $3 \times 3$ MEC example

For two finite marginals, Shannon entropy is concave on the transportation polytope, so a minimum occurs at an extreme point. For the  $3 \times 3$  case an extreme point has at most five positive cells. One may therefore enumerate all five-cell supports, solve the independent row/column constraints, retain nonnegative feasible solutions, and evaluate the joint entropy. This finite enumeration yields (8.2).

For the specific  $q$  in (8.1), the nonzero entries of (8.2) are

$$\frac{1}{2} - q, \quad q, \quad 2q - \frac{3}{4}, \quad 1 - 2q, \quad \frac{1}{4},$$

all nonnegative because  $q \in (3/8, 1/2)$ . Their entropy is the value in (8.3).

## B Why equality at the entropy floor forces invertibility

Let  $X$  be a finite random variable and  $Y = f(X)$  deterministic. Then

$$H(X) = H(Y) + H(X | Y).$$

Thus

$$H(X) = H(Y)$$

if and only if

$$H(X | Y) = 0,$$

which is equivalent to  $X$  being an almost-sure function of  $Y$ . Hence  $f$  is invertible on the positive-probability atoms after identifying zero-probability states. This elementary equality condition is the engine behind the spectrum-cancellation theorem.

## C Literature map rather than novelty claim

Table 1 records the present division between known neighboring theory and questions that remain open *in the specific formulation of this note*. “Open here” means only that this note does not know the answer; it is not a historical priority claim.

| Topic                                                | Status      | Representative reference                        |
|------------------------------------------------------|-------------|-------------------------------------------------|
| Minimum-entropy coupling                             | Known       | Kovačević–Stanojević–Šenk [2]                   |
| Exact random-number generation                       | Known       | Han–Hoshi [3]; Watanabe–Han [4]                 |
| Online randomness recycling                          | Known       | Draper–Saad [5]                                 |
| Near-Shannon online sampling with real probabilities | Known       | Draper–Harris–Saad [6]                          |
| Space lower bounds near Shannon rate                 | Known       | Draper–Saad [7]                                 |
| Mixed/behavioral realization and finite memory       | Known       | Kuhn [8]; Koller–Megiddo [9]; Main–Randour [10] |
| Joint counterfactual action coupling in MDPs         | Known       | Kaya–Ghasemi–Hashemi [11]                       |
| Stable exogenous event identity across interventions | Known issue | Buffalo–Pearson–Klein [12]                      |
| Correlated/reversible catalysis and entropy          | Known       | Wilming [13, 14]                                |
| Exact counterfactual memory–innovation frontier      | Open here   | No direct source located in this formulation    |
| Full-tree vs chosen-path entropy-rate gap            | Open here   | No direct source located in this formulation    |
| Finite-memory decay of spectral surcharge            | Open here   | No direct source located in this formulation    |

Table 1: A deliberately conservative map of the neighborhood.

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